

Further we moved over to programming the Arduino with a small note on how to upload your code to the Arduino. We covered the basic operations that can be performed by the Arduino and gave some references to help you with making some. In the context of this FastTrack to Internet of Things, we covered the part of measuring the values from the sensors and the also passing the data to the Internet.

Raspberry Pi?

Though the name sounds as it being something that you can eat just right away, we won't suggest you to eat it, but you can instead use to monitor the process of making a Raspberry Pi.

The Raspberry is an ARM cortex A series processor based hardware device. It has the capability of a modern cell phone and is therefore more powerful than the Arduino boards available in the market as of this writing. It runs an operating system called Raspbian which is a highly customized Linux flavour for the Raspberry Pi.

It can be programmed using a visual programming language called Scratch. Though for building the Internet of Things, you'll have to know a more advanced programming language like Python or C, for handling all the Internet communication requests.

How can Arduino and Raspberry Pi help me build the Internet of Things?

As we previously mentioned they can be used to create the 'things' in the Internet of Things. These 'things' can measure value from sensors or help control some device or hardware.

Part 1 Measurements

The preliminary task of a 'thing' in the Internet of Things is to collect data of various phenomenon around us like temperature, moisture for adjusting room temperature. This represents measurement and plays a crucial part in the IoT phenomenon.

In the Fast Track to the Arduino for everyone, we have used demonstrated how the Arduino can be used

- ◆ To get current position using GPS
- ◆ To measure distance using Ultrasonic sensor
- ◆ To interface an accelerometer
- ◆ To interface a Gyroscope